PLAN — Depth-dependent stress magnitudes for the CSM (YHSM-2013) fault projection

2026-06-15 — rev 7

Contents

- Date: 2026-06-15 (rev 7)
- Status: PLAN (no implementation yet; current tool implements constant prescribed principals 80/40/35 MPa). Four of the open decisions are now FIXED by user directive 2026-06-15 (density = MUSCAL, pore pressure = hydrostatic, closure = Luttrell differential stress, axes/taper unchanged) — see Section 8.
- Tool: `project_csm_stress_to_vtu.py` (this folder)
- Data (orientations): `raw_data/yang_and_hauksson/CSM_data_1781290479811.csv` (YHSM-2013, orientation only)
- Data (differential stress, NEW rev 5): `raw_data/luttrell_differential_stress/CSM_data_1781537253009.` (Luttrell-2017 CSM export; D = S1-S3 column, see Section 3 closure C2-s)
- Mesh: `safs_seisol_v2_0_0_RSSRW/safs_mesh.puml.h5` (UTM 11N, fault z from -21 m to -16,528 m)
- Revision history:
  - rev 7 (2026-06-15) adds the `--sample {nearest,linear}` orientation sampler (Section 5). Nearest-neighbour sampling of the ~2 km CSM grid made the mu / tau field blocky (60,658 facets -> ~719 distinct CSM cells); `--sample linear` barycentrically interpolates the CSM tensor components per facet for a smooth orientation field (NN fallback outside the data hull). Default stays `nearest` (byte-identical to prior runs); the mu cap is preserved under either.
  - rev 6 (2026-06-15) replaces the closure-C2 positivity cap. The rev-5 implementation capped sig3 to a fixed fraction of Sv_eff, which pinned the stress ratio sig1/sig3 at ~30

on shallow facets (super-critical) and produced unphysical apparent friction mu -> 1.0 at the free surface. The cap is now a FRICTIONAL (Mohr-Coulomb / Byerlee) ceiling: D_applied = min(D, D_max) with D_max = (k_max-1)*Sv_eff/((1-R)*k_max+R), bounding sig1/sig3 <= k_max (CLI `--cap-k-max`, default 5.83 = mu 1.0) and tapering the differential to 0 as Sv_eff -> 0 at the surface. Adds references 23 (Sibson 1974), 24 (Brace & Kohlstedt 1980), 25 (Townend & Zoback 2000); see Sections 3 (C2-s cap), 4, 7. The seismogenic-depth field and the hypocenter are unchanged (those facets were never capped).

- rev 5 (2026-06-15) records four user decisions and the arrival of the Luttrell spatial differential-stress data: (1) DENSITY = the MUSCAL lateral-mean rho(z) profile is the fixed default; the Vp->rho-recipe confirmation is downgraded from a blocker to a manuscript footnote (Section 2.1, decisions 5+7). (2) PORE PRESSURE = hydrostatic is the fixed default (Section 2.2, decision 2).
  (3) CLOSURE = the differential stress D = S1-S3 from Luttrell & Smith-Konter (2017) is now the recommended default, used SPATIALLY (closure C2-s) from the CSM-portal export that has been obtained and placed in `raw_data/luttrell_differential_stress/` — this resolves decisions 1 and 6 and flips the rev-3 recommendation away from the frictional ratio C1 (Sections 0, 3, 4, 6). (4) The C1-vs-C2 comparison is expanded into a dedicated subsection (Section 3.x) and the worked example is rerun across depths with the on-fault Luttrell field (Section 4). The frictional ratio C1 is retained as the `--closure ratio` alternative.
- rev 4 corrects the density provenance: the project's material file derives from the raw extracts in `velocity/raw/multiscale_statewise_cvm/` — the SCEC CVM Explorer model "muscal" (Multi-scale Statewide CALifornia model, Yeh & Ben-Zion, CANVAS base) — NOT CVM-H, and rev 3's claim that the densities are Nafe-Drake/Brocher conversions is downgraded to "conventional expectation, to be confirmed against the muscal publication" (Section 2.1).
- rev 3 addresses four review comments: (1) verified the CVM density variable and clarified bulk-vs-dry density for the lithostatic integral (Section 2.1); (2) replaced the bare Hubbert-Rubey justification with SAF-specific fluid-pressure measurements and models (Section 2.2); (3) removed all Parkfield/SAFOD-derived values — creeping central SAF, not representative of the locked southern SAF (Sections 3, 6, 7); (4) added a path to using the Luttrell & Smith-Konter constraint as a SPATIAL field (Section 3, closure C2-s).
- rev 2 removed every reference to, and every parameter value taken from, the Kaikoura (New Zealand) dynamic rupture literature (Ulrich et al. 2019); those are NOT San Andreas constraints.

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0. Goal, approach, and scope rule

The CSM model YHSM-2013 (Yang & Hauksson 2013) provides stress ORIENTATION only — its tensors are normalized deviatoric ("This model provides orientation only. Magnitude information is not meaningful.", csv header). The current tool therefore prescribes three constant effective principal magnitudes. This plan upgrades the magnitudes to a depth-dependent, observation-constrained model in three steps:

1. S2 = effective vertical stress Sv_eff(z) = total overburden (lithostatic, BULK density from MUSCAL) minus hydrostatic pore pressure — Step 1.
2. A literature closure linking the two horizontal principals S1, S3. The DEFAULT (rev 5) is the differential stress D = S1 - S3 taken as a SPATIAL field D(x,y) from Luttrell & Smith-Konter (2017) — closure C2-s. The frictional effective ratio k = S1/S3 (closure C1) is kept as an alternative — Step 2.
3. Combine Sv_eff(z), the closure, and the PER-POINT stress shape ratio R from the CSM csv (column 13) to solve for all three magnitudes S1(x,z) >= S2(z) >= S3(x,z) in closed form — Step 3.

Then rebuild the regional tensor from the CSM angles (principal-axis eigenvectors / SHmax azimuth, csv columns 10, 21–35) and project onto the fault with the existing machinery — Steps 4–5.

SCOPE RULE (standing constraint for this work). This is a San Andreas Fault model. Every parameter VALUE must trace to (a) observations in the SAF system / southern California, or (b) setting-independent mechanics (effective-stress principle, Mohr-Coulomb friction, Anderson faulting theory). Two exclusions:

- Values calibrated for OTHER earthquakes or regions (Kaikoura, Landers-event specifics, subduction settings, …) must not be imported, however similar the methodology looks.
- Parkfield / SAFOD data (pilot-hole stress, SAFOD core gouge friction) are EXCLUDED from value-carrying roles: Parkfield sits in the creeping central SAF, mechanically unlike the locked southern SAF this mesh models (user directive 2026-06-12).

Methodology-only citations are allowed but must be flagged as such.

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1. Conventions – READ FIRST (sign/ratio traps)

All formulas below use SEAS compression-positive effective principal stresses, written

```
sig1 >= sig2 >= sig3 > 0     (compression positive, EFFECTIVE, MPa)
```

The csv is TENSION-positive ("Tension is taken as positive"), with S1 = most tensional. Mapping between the two:

```
sig1 = -S3_csv   (csv most COMPRESSIONAL principal -> our largest)
sig2 = -S2_csv
sig3 = -S1_csv   (csv most TENSIONAL principal -> our smallest)
```

Shape-ratio dictionary (verified numerically on all 22,131 csv rows, agreement to 5e-4 = csv rounding):

```
csv col 13:  R_GF  = (S2-S3)/(S1-S3)   tension-positive
                   = (sig1-sig2)/(sig1-sig3)   compression-positive

csv col 12:  phi   = (S1-S2)/(S1-S3)   tension-positive  (Angelier 1979)
                   = (sig2-sig3)/(sig1-sig3)   compression-positive
```

```
                      (this compression-positive form is often written
                       "nu" or "Phi" in the dynamic-rupture literature)

identity:    phi + R_GF = 1   (holds exactly in the csv)
```

So: when this plan says "R from the csv" it means column 13. Do not mix the two columns.

Useful interpolation forms (both equivalent):

```
sig2 = (1 - R)*sig1 + R*sig3            with R   = csv col 13
sig2 = phi*sig1 + (1 - phi)*sig3        with phi = csv col 12
```

Frame: (east, north, up) identified with mesh (x, y, z); UTM Zone 11N metres; lon/lat -> UTM via pyproj EPSG:4326 -> EPSG:32611, always_xy (velocity-pipeline convention). Grid convergence (< ~1 deg) neglected.

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2. Step 1 – S2 = effective vertical stress Sv_eff(z)

```
Sv_total(z) = integral_0^z rho_bulk(z') * g * dz'    (total overburden)
P_p(z)      = pore pressure model (Section 2.2)
Sv_eff(z)   = Sv_total(z) - P_p(z)
```

2.1 Density: what is in the MUSCAL material file, and bulk vs dry

Do we have density? Yes — verified 2026-06-12. `safs_seisol_v2_0_0_RSSRW/safs_material_cvm.nc` stores a compound variable `data{rho, mu, lambda}` on a 264 x 195 x 181 UTM grid (dz = 250 m, z = 0 .. -45 km, units attribute "rho kg/m^3"). The `rho` field is carried through unmodified from the raw extracts in `velocity/raw/multiscale_statewide_cvm/` (`Lon,Lat,Vp(m/s),Vs(m/s), Density(kg/m^3)` columns; see `velocity/code/raw_readers.py` and `toolbox/generate_velocity_nc_from_raw`).

Which velocity model is this? The raw slice headers say `# CVM(abbr): muscal` — the SCEC CVM Explorer export of MUSCAL, the Multi-scale Statewide CALifornia velocity model (Yeh, Ben-Zion & Olsen, ESSOAr preprint 2026; UCVM >= 25.7 component `SCECcode/muscal`, authors file: Te-Yang Yeh and Yehuda Ben-Zion, USC). MUSCAL builds on the CANVAS adjoint-waveform-tomography base model of California and Nevada (Doody et al. 2023) and merges regional/local datasets with a near-surface low-velocity taper. Note this is NOT CVM-H; the `velocity/README.md` header line predates the switch (three raw versions exist — `cvmh/`, `cvm_s4.26.m01/`, `multiscale_statewide_cvm/` — and the production sidecar recorded in the nc attribute `source_sidecar` is the muscal one).

Computed lateral-mean profile (this mesh region):

```
z =      0 m   mean rho = 1678 kg/m^3   (basin sediments at surface)
z =  -1000 m   mean rho = 2450
z =  -2000 m   mean rho = 2525
z =  -5000 m   mean rho = 2669
z = -10000 m   mean rho = 2773
z = -15000 m   mean rho = 2759
```

```
global range 1245 .. 3341 kg/m^3
```

Is it rock density or rock+fluid combined? It is in-situ BULK density (rock matrix + pore fluid), with one caveat on provenance. The `rho` array ships inside the authors' own model file (`model_MUSCAL_CANVAS_dll0.01_vardz_float32_cmpd.nc`; the UCVM packaging only re-encodes it — verified in `SCECcode/muscal data/rewrite2bin.py`). Seismic tomography does not invert for density, so MUSCAL's rho is necessarily ASSIGNED from the velocities; the universal practice is an empirical Vp-density relation (Nafe-Drake curve as formulated by Brocher 2005), whose calibration data are saturated, in-situ rocks — i.e. bulk density. The exact relation MUSCAL used is not stated in the UCVM packaging. The shallow values (lateral mean 1678 kg/m^3 at the surface, minimum 1245) are only consistent with saturated-sediment bulk densities, supporting this reading.

DECISION (user directive 2026-06-15, resolves decisions 5+7): USE the MUSCAL density. The rho field shipped in `safs_material_cvm.nc` is taken AS-IS — its lateral-mean rho(z) profile (table below) is the fixed default for the lithostatic integral. The Vp->rho-recipe confirmation that rev 3/4 listed as a pre-implementation ACTION is DOWNGRADED to a non-blocking manuscript footnote: cite the bulk-density reading as the conventional expectation (Brocher 2005), to be replaced by the model's own documentation when the Yeh, Ben-Zion & Olsen preprint or software@scec.org confirms the exact relation. It does not gate implementation.

And that is the CORRECT density for this integral. The total vertical stress at depth is the weight of EVERYTHING above — mineral grains and the fluid filling their pores:

```
Sv_total(z) = integral rho_bulk * g * dz      (bulk = correct)
```

Using a dry-skeleton density would UNDERESTIMATE the overburden. The "dry rock minus fluid" intuition from the original task statement is realized by the effective-stress subtraction Sv_eff = Sv_total - P_p (Hubbert & Rubey 1959), NOT by integrating a dry density. So: bulk density in the integral, pore pressure subtracted after — no double counting.

Magnitude check (lateral-mean MUSCAL profile vs constant 2670 kg/m^3):

```
z = -5.0  km: Sv_cvm = 123.6 MPa   const-2670 = 131.0   (-5.7 %)
z = -10.0 km: Sv_cvm = 258.3 MPa   const-2670 = 261.9   (-1.4 %)
z = -16.5 km: Sv_cvm = 434.4 MPa   const-2670 = 432.2   (+0.5 %)
```

The difference is largest in the upper 5 km (low-density basin sediments — Coachella/Salton region), exactly where the fault mesh is shallow; this is why the MUSCAL profile is the DEFAULT over a constant. Implement as `--density-source {muscal,constant}` (default `muscal`) with `--material-nc PATH` (the production file is `safs_seisol_v2_0_0_RSSRW/safs_material_cvm.nc`; the name retains the "cvm" tag but the contents are MUSCAL, see above). `constant` (with `--rho-const-kgm3`, default 2670) is kept only for the magnitude check and reproducibility. Use the laterally-AVERAGED rho(z) profile, not per-column integration: per-column Sv would introduce lateral Sv jumps that are not in mechanical equilibrium and would leak into mu maps.

2.2 Pore pressure: SAF-specific observations and models

What Step 1 needs is P_p(z) in the REGIONAL field (country rock) — the fault-core fluid state is a separate, local matter. The available SAF-system constraints (searched 2026-06-12; there is NO community pore-pressure model for southern California analogous to CVM/CFM/CSM):

- DIRECT measurement, southern California, 4 km from the SAF: Cajon Pass scientific borehole. In situ fluid-pressure build-up tests at ∼1.8–2.1 km depth give pressures only ∼5 % above hydrostatic (Coyle & Zoback 1988). Independently, the measured stress magnitudes over 0.9–3.5 km are consistent with frictional equilibrium at mu = 0.6–1.0 UNDER HYDROSTATIC pore pressure (Zoback & Healy 1992).
- SAF-specific fluid-pressure DISTRIBUTION models: Rice (1992) (idealized) and Fulton & Saffer (2009) (numerical, fed by the mantle-helium fluid source of Kennedy et al. 1997) both conclude that overpressure develops INSIDE the low-permeability fault zone while the surrounding crust remains near-hydrostatic. Fulton & Saffer quantify the fault-zone EXCESS pressure: ∼1 MPa at 2.7 km, ∼17 MPa at 6 km depth, confined to the fault zone and mostly below 5 km.

DECISION (user directive 2026-06-15, resolves decision 2): USE hydrostatic pore pressure for the regional field. This is the defensible model and is the fixed default,

```
P_p(z) = rho_w * g * z          (rho_w = 1000 kg/m^3)
lambda(z) = P_p/Sv_total = 0.37 - 0.40 for the MUSCAL profile above
            (0.397 at 5 km, 0.380 at 10 km, 0.373 at 16.5 km)
```

CLI: `--pp-model hydrostatic` (default). A FAULT-LOCAL effective normal stress reduction following the Fulton & Saffer (2009) excess- pressure depth profile is a possible later refinement — it would be applied on the fault surface after projection, not to the regional tensor; out of scope for this tool, noted for the friction setup.

(Hubbert & Rubey 1959 is cited only for the effective-stress principle and the lambda notation — methodology, not values.)

2.3 Optional deep taper of the deviatoric part

Below the seismogenic zone the crust creeps and cannot sustain large deviatoric stress, so the deviatoric part should taper out with depth while the isotropic effective part continues. SAF-specific depth scale: the southern California seismogenic thickness is 15.0 (+1.2/ -1.1) km on average, locally < 10 km in the Salton Trough and > 20 km elsewhere (Nazareth & Hauksson 2004) — directly relevant since the SAFS mesh spans the Coachella/San Gorgonio corridor and reaches z = -16.5 km.

```
sigma_eff(x,z) = Omega(z) * sigma_dev_eff(x,z) + Sv_iso_eff(z) * I
Omega(z) = 1 above z_seis, smooth -> 0 below     (default OFF)
--taper-zseis-km 15      (Nazareth & Hauksson 2004 regional average)
```

Keep OFF by default until first dynamic-rupture trial; the fault mesh bottom (16.5 km) is near the regional seismogenic floor anyway.

2.4 Regime validity check (S2 = vertical assumption)

Assigning the vertical stress to S2 assumes a strike-slip (Andersonian) regime, Aphi in [1, 2] (Anderson faulting classes; Aphi of Simpson 1997). Measured ON THIS FAULT MESH by nearest-neighbour sampling of the csv (computed 2026-06-12):

```
Aphi < 1 (normal-transtensional)   16.8 % of facets
1 <= Aphi <= 2 (strike-slip)       76.2 %
Aphi > 2 (reverse-transpressive)    7.0 %    (San Gorgonio corridor)

V2 (intermediate axis) plunge:  median 73.6 deg
  plunge >= 60 deg: 67.8 %   plunge >= 45 deg: 76.2 %
```

So S2 ~ vertical is a good approximation on ~3/4 of the fault and questionable on the rest (consistent with the transpressive San Gorgonio knot in the regional studies, Yang & Hauksson 2013; Hardebeck & Hauksson 2001). Mitigation: output the sampled V2 plunge and Aphi as cell fields (`csm_V2_plunge_deg_cell`, `csm_Aphi_cell`) so the violation zones are visible in ParaView, and report the sigma_zz_eff vs Sv_eff(z) discrepancy statistics in the summary (Section 6).

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3. Step 2 – Literature closure between S1 and S3 (SAF region)

One more scalar relation is needed to close the system. Two usable forms, both to be implemented. ALL supporting values below are from the SAF system / southern California; Parkfield/SAFOD data are excluded per the scope rule.

Closure C1 – effective stress ratio k = sig1/sig3 (ALTERNATIVE, rev 5)

Frictional-equilibrium limit on optimally oriented faults (Mohr-Coulomb; friction range mu = 0.6–1.0 of Byerlee 1978):

```
k = sig1/sig3 = ( sqrt(mu^2 + 1) + mu )^2
mu = 0.6  ->  k = 3.12
mu = 1.0  ->  k = 5.83
```

SAF-region evidence:

- Cajon Pass borehole, 4.2 km from the SAF, southern California (Zoback & Healy 1992): measured shear/normal stress ratios on favorably oriented planes at 0.9–3.5 km depth match mu = 0.6–1.0 with hydrostatic pore pressure. This is the closest direct magnitude measurement to the modeled fault section and the primary justification for k ~ 3 (mu = 0.6, lower bound of the measured range) in the REGIONAL field.
- Townend & Zoback (2004): stress fields adjacent to the SAF in central and southern California are consistent with a STRONG crust while the SAF itself slips at high SHmax angle (up to ~85 deg) — "strong crust, weak fault". The CSM ORIENTATIONS supply the misorientation; k supplies the strong-crust magnitude.
- Strong-fault counterpoint: Scholz (2000) argues the heat-flow evidence is not decisive and the SAF may sustain Byerlee-level stress. Either way k ~ 3 is the defensible regional value; the weak-vs-strong debate concerns the resolved stress ON the fault, which here comes from the CSM orientations.

Closure C2 – differential stress D = sig1 - sig3 (DEFAULT, rev 5)

Luttrell & Smith-Konter (2017), from southern California topography + focal mechanisms: differential stress at seismogenic depth must EXCEED ~20 MPa in most of southern California and ~62

MPa near the most rugged SAF topography (which includes San Gorgonio, on this mesh). As of rev 5 their model has been obtained as a gridded field (closure C2-s below), so D is used SPATIALLY rather than as a single dialed scalar. Use D as a direct dial:

```
D options: spatial D(x,y) from the Luttrell field (DEFAULT, C2-s);
           or a constant scalar D (e.g. 30-60 MPa, --diff-MPa);
           or D proportional to Sv_eff(z) (self-similar with depth;
           equivalent to a fixed k).
```

A note on depth dependence. The Luttrell field is published at a single reference depth (5 km below sea level) and the authors state explicitly that "these model values do not vary with depth" (csv header). So in closure C2 the differential stress D(x) is held CONSTANT with depth while the intermediate principal sig2 = Sv_eff(z) grows with depth — i.e. the deviatoric part is BOUNDED and the overburden carries the depth dependence. This is the defining contrast with closure C1 (where D grows linearly with depth); see Section 3.x.

Closure C2-s – SPATIAL Luttrell & Smith-Konter field (OBTAINED, rev 5)

Review question (rev 3): "can we get the spatial distribution of this?" Answer (rev 5): YES — the data is now in hand. The Luttrell & Smith-Konter (2017) model is on the SAME SCEC Community Stress Model portal that this project's YHSM-2013 csv came from, exported in the identical `CSM_data_*.csv` 35-column format, and has been placed at:

```
raw_data/luttrell_differential_stress/CSM_data_1781537253009.csv
  header: CSM model "Luttrell-2017", contributed by K. Luttrell and
          B. Smith-Konter; "provides both meaningful orientation and
          meaningful magnitude"; tension-positive MPa, WGS84 lon/lat.
  19,100 valid rows on a ~0.02 deg (~2 km) lon/lat grid
          (lon -119.149 .. -114.918, lat 32.625 .. 35.252).
  single depth = 5 km below sea level; "values do not vary with depth".
  column 16 (1-based) = "differential stress (S1-S3) in MPa".  <-- USE
  (column 15 "isotropic pressure" is EMPTY for every row: this is a
   deviatoric / differential-stress model, no absolute mean stress.)
```

The grid FULLY contains the SAFS fault footprint (fault lon -118.474 .. -115.686, lat 33.365 .. 34.684), so every facet has a nearby sample. Sampling the field over the fault footprint (2026-06-15, this folder):

```
on-fault D = S1-S3 (MPa):  min ~0.1    p25 ~12    median ~19    mean ~20
                           p75 ~26     p90 ~34     max ~60
  hypocenter facet (Coachella/Salton end):  D ~ 12 MPa
  San Gorgonio knot (rugged SAF topography): up to ~60 MPa
```

These reproduce the paper's headline bounds (>= ~20 MPa regionally, ~62 MPa at the most rugged SAF topography) directly on the mesh.

Use of D(x) (= column 16) with a SECOND nearest-neighbour sampler, identical in form to the CSM-orientation sampler (depth ignored — the field is depth-invariant by construction):

```
read:    parse the same way as the YHSM csv (lon, lat, col 16 -> D);
         drop rows with NaN D.
sample:  build a cKDTree on the Luttrell (x,y) in UTM 11N; query at
         each fault-facet centroid (x,y); D_facet = D at nearest node.
apply:   closure C2 (Section 4) with this per-facet D, depth-constant:
             sig1 = Sv_eff(z) + R(x)*D(x)
             sig3 = Sv_eff(z) - (1-R(x))*D(x)
cap:     FRICTIONAL ceiling (rev 6). The brittle crust cannot sustain a
         differential stress above the Mohr-Coulomb/Byerlee frictional
         strength, so the stress ratio sig1/sig3 is bounded by
         k_max = (sqrt(mu^2+1)+mu)^2; expressed as a differential cap
         this is the closure-C1 value at k_max:
             D_max(z) = (k_max - 1)*Sv_eff(z) / ((1-R)*k_max + R)
             D_applied = min(D(x), D_max(z))
         Because D_max is PROPORTIONAL to Sv_eff it -> 0 at the free
         surface, so the near-surface differential vanishes instead of
         producing a super-critical ratio. Count and report capped
         facets. (Byerlee 1978; Sibson 1974; Brace & Kohlstedt 1980;
         observed in the SAF region by Zoback & Healy 1992 and Townend &
         Zoback 2000.)
CLI:     --closure diff  --diff-csv PATH  --cap-k-max 5.83  (PATH
         defaults to the Luttrell file above; --diff-MPa is the scalar
         fallback when no csv is given; --cap-k-max default 5.83 = mu 1.0
         Byerlee ceiling, use 3.12 for mu 0.6).
```

Validation against the published bound becomes a self-consistency check: D_applied(x) == D_Luttrell(x) on every UN-capped facet, and the capped-facet count quantifies where the frictional ceiling bit (the shallow + high-D San Gorgonio facets).

3.x – C1 vs C2: how the two closures differ (and why C2 is default)

Both closures take sig2 = Sv_eff(z) as the (sub-)vertical intermediate principal and use the SAME per-point shape ratio R to place sig2 between sig1 and sig3. They differ ONLY in what sets the SPREAD of sig1 and sig3 around sig2 — and that single difference propagates into opposite depth behaviour and opposite stress levels.

1) What sets the magnitude scale.

```
C1 (ratio):  sig1 = k * sig3,  k = ((mu^2+1)^0.5 + mu)^2 fixed scalar.
             The scale is tied to FRICTION (Mohr-Coulomb equilibrium on
             an optimally oriented fault).  No magnitude data needed.
C2 (diff):   D = sig1 - sig3 prescribed directly (here = Luttrell(x)).
             The scale is tied to an OBSERVATIONAL differential-stress
             bound (topography + focal mechanisms).  Needs a D field.
```

2) Depth behaviour — the key divergence. Because sig3 (C1) is proportional to Sv_eff(z), the C1 differential stress GROWS with depth; the C2 differential stress is HELD CONSTANT (Luttrell

is depth- invariant). Writing both as a function of Sv_eff:

```
C1:  D_C1(z) = sig1 − sig3 = (k−1) ∗ sig3
             = (k−1) / ((1−R)∗k + R) ∗ Sv_eff(z)     ~ proportional to z
     max shear (sig1−sig3)/2 grows without bound with depth.
C2:  D_C2    = Luttrell(x)  = constant in z
     max shear = D/2 is the SAME at every depth (a few tens of MPa).
```

3) Numbers on this mesh (MUSCAL rho, hydrostatic P_p, on-fault median R = 0.44; C1 with k = 3.12; C2 with the on-fault median Luttrell D = 19 MPa):

```
depth   Sv_eff   C1 max−shear (k=3.12)   C2 max−shear (D=19)
 5 km    74.6 MPa      36.1 MPa                 9.5 MPa
10 km   160.2 MPa      77.7 MPa                 9.5 MPa
16.5 km 272.5 MPa     132.0 MPa                 9.5 MPa
```

C1 climbs to >100 MPa of shear at the base of the seismogenic zone ("strong crust", Byerlee-bounded). C2 stays flat at D/2 — ∼9.5 MPa for a median facet, up to ∼30 MPa for the San Gorgonio high-D (D∼60) facets — i.e. at or below the ∼20 MPa-class heat-flow / weak-fault bound (Lachenbruch & Sass 1980, 1992) almost everywhere.

4) Consequences for the rupture model.

```
− Resolved shear tau and mu_apparent scale with the deviatoric part,
  so C1 produces MUCH higher tau / mu_apparent than C2 (the constant−
  magnitude run, ~C1 at 4−5 km, already sits at mu_app median 0.34).
  C2/Luttrell brings mu_apparent toward the 0.1−0.3 weak−fault target.
− C1 needs no spatial data but cannot represent lateral contrasts;
  C2−s carries the San Gorgonio differential−stress high directly from
  the data (~60 MPa there vs ~12 MPa at the Coachella/Salton hypocenter).
− C2's positivity limit D < Sv_eff/(1−R) bites at shallow depth where
  D is large (capping needed); C1 is automatically ordered for k>=1.
```

5) Why C2/Luttrell is the rev-5 default. The modelled fault is the LOCKED southern SAF, where the heat-flow and stress-rotation observations (Sections 3 consistency check, 6) point to a weak fault in a moderately stressed crust — a BOUNDED deviatoric stress of a few tens of MPa, exactly what the Luttrell field supplies, and spatially resolved. C1 (frictional, strong-crust) is retained as `--closure ratio` for sensitivity studies and the strong-crust counterpoint (Scholz 2000).

Consistency check against SAF weak-fault observations

Whatever closure is chosen, the RESOLVED stress on the SAF must be checked after projection (Section 6) against:

- heat-flow bound: fault-parallel shear stress averaged over the seismogenic layer <= ∼20 MPa (Lachenbruch & Sass 1980, SAF-wide; Lachenbruch & Sass 1992 specifically at Cajon Pass; synthesis Zoback et al. 1987);
- low apparent friction on the SAF (∼0.1–0.3) implied by the high SHmax-to-fault angle in the

southern California inversions (Hardebeck & Hauksson 2001; Townend & Zoback 2004).

If the projected tau / mu_apparent grossly exceed these, reduce k (or D) toward the weak end — with the reduction documented against these same references.

Recommended default (rev 5, user directive): C2-s — the SPATIAL Luttrell differential-stress field. CLI:

```
DEFAULT:  --closure diff   --diff-csv <LUTTRELL_CSV>   (spatial, C2-s)
fallback: --closure diff   --diff-MPa 40               (scalar, no csv)
alt:      --closure ratio  --k-ratio 3.12              (frictional C1)
  <LUTTRELL_CSV> = raw_data/luttrell_differential_stress/
                   CSM_data_1781537253009.csv
```

---

4. Step 3 – Closed-form magnitudes from (Sv_eff, closure, R)

Given sig2 = Sv_eff(z) and the csv R = R_GF (column 13) at the sampled CSM point:

With C1 (ratio k):

```
sig2 = (1-R)*sig1 + R*sig3   and   sig1 = k*sig3
=>  sig3(x,z) = Sv_eff(z) / ( (1-R(x))*k + R(x) )
    sig1(x,z) = k * sig3(x,z)
```

Ordering is automatic for k >= 1 and R in [0,1]: sig1 >= sig2 <=> R_(k-1) >= 0; sig2 >= sig3 <=> (1-R)_(k-1) >= 0.

With C2 (differential D) – DEFAULT, rev 5:

```
sig1(x,z) = Sv_eff(z) + R(x)*D(x)
sig3(x,z) = Sv_eff(z) - (1-R(x))*D(x)
D(x) = Luttrell(x) (closure C2-s, depth-constant), or scalar --diff-MPa
frictional cap:  D_applied = min(D, D_max),
                 D_max(z) = (k_max-1)*Sv_eff(z) / ((1-R)*k_max + R)
                 => sig1/sig3 <= k_max  (Byerlee ceiling, default 5.83)
```

The naive positivity bound D < Sv_eff/(1-R) is necessary but NOT sufficient: it keeps sig3 > 0 yet still admits a near-zero sig3, i.e. a super-critical sig1/sig3 ratio that produces unphysical apparent friction (mu -> 1.0) at the free surface. The FRICTIONAL cap above (rev 6) is the physical one: the brittle crust cannot exceed Mohr-Coulomb/Byerlee frictional strength, so sig1/sig3 <= k_max, and since D_max is proportional to Sv_eff the differential vanishes as the overburden -> 0 at the surface (Byerlee 1978; Sibson 1974; Brace & Kohlstedt 1980; Zoback & Healy 1992; Townend & Zoback 2000). This bites at shallow depth and where the Luttrell D is large (San Gorgonio high); count and report capped facets.

Worked example (on-fault median R = 0.44, MUSCAL profile, hydrostatic P_p):

```
                       z = 5 km    z = 10 km   z = 16.5 km
Sv_total               123.6 MPa   258.3 MPa   434.4 MPa   (MUSCAL, Sec 2.1)
P_p (hydrostatic)       49.1 MPa    98.1 MPa   161.9 MPa
Sv_eff = sig2           74.6 MPa   160.2 MPa   272.5 MPa

C1, k = 3.12 (frictional, strong crust):
  sig3                  34.1 MPa    73.2 MPa   124.6 MPa
  sig1                 106.3 MPa   228.5 MPa   388.7 MPa
  max shear             36.1 MPa    77.7 MPa   132.0 MPa  <- grows with z

C2-s, D = Luttrell (default); median on-fault D = 19 MPa, depth-const:
  sig1                  82.9 MPa   168.6 MPa   280.9 MPa
  sig3                  63.9 MPa   149.6 MPa   261.9 MPa
  max shear              9.5 MPa     9.5 MPa     9.5 MPa  <- flat (= D/2)
  (San Gorgonio facet, D ~ 60 MPa: max shear ~30 MPa, capped if shallow)
```

The depth divergence is the whole point: C1 max shear climbs 36 -> 132 MPa over the seismogenic column, while C2/Luttrell stays flat at D/2. For comparison the current constant-magnitude run uses 80/40/35 MPa at ALL depths — roughly the C1 numbers at z ∼ 4–5 km. The depth- dependent model removes exactly this inconsistency, and the C2/Luttrell default keeps the resolved shear within the SAF weak-fault bounds (Section 3 consistency check).

---

5. Step 4 – Assemble the tensor from the csv angles

Per CSM point we have the principal axes; two assembly options:

Option A (recommended default; matches current tool + user intent)

Use the full eigenvectors of the csv tensor (columns 21–29, or equivalently our own eigh of See..Suu — already verified identical, |dot| median 1.0000):

```
sigma_eff(x,z) = sig1 * u_c u_c^T + sig2 * u_i u_i^T + sig3 * u_t u_t^T
u_c = most-compressional axis (csv V3)
u_i = intermediate axis        (csv V2)
u_t = most-tensional axis      (csv V1)
```

Honest caveat: sig2 = Sv_eff(z) is assigned to u_i, which is only SUB-vertical (median plunge 73.6 deg on the fault); the built tensor's sigma_zz_eff will deviate from Sv_eff(z) where the axes tilt. Report the deviation stats; flag low-plunge zones via cell fields. The benefit: the measured transpressive/transtensional style (San Gorgonio vs Salton) is retained.

Option B (strict Andersonian)

Anderson faulting theory (Anderson 1951): one principal axis exactly vertical. Use only the SHmax azimuth (csv column 10):

```
e_H = (sin(az), cos(az), 0)    az = SHmax deg E of N
e_h = (cos(az), -sin(az), 0)   (horizontal, perpendicular)
e_v = (0, 0, 1)
sigma_eff = sig1 * e_H e_H^T + sig3 * e_h e_h^T + sig2 * e_v e_v^T
```

Cleaner mechanics (sigma_zz_eff == Sv_eff exactly), but discards the measured axis plunges — in the 24% non-strike-slip zones the local style (e.g. the San Gorgonio thrust component) is lost. Implement both: `--axes {csm,andersonian}`.

Either way the per-point tensor then goes through the EXISTING pipeline: CSM-orientation sampling at facet centroids (now also using the facet centroid DEPTH for Sv_eff), Tandem basis (s = up x n, d = s x n), normal harmonisation to the SW half-space, traction resolution, mu_apparent = tau_mag / sigma_n_eff.

Orientation sampling — nearest vs linear (rev 7). The CSM (YHSM-2013) grid is ~2 km (0.02 deg); the fault mesh is ~430 m triangles, so nearest-neighbour sampling makes the orientation (and shape ratio R) PIECEWISE-CONSTANT in the ~2 km CSM cells: on this mesh 60,658 facets map to ~719 distinct CSM points (median ~78 facets per cell), which renders as blocky mu / tau patches. `--sample linear` instead does a barycentric (Delaunay) interpolation of the SIX tension-positive tensor components onto each facet centroid and re-eigendecomposes per facet to get a SMOOTH axis + R field (SHmax / eig-gap / Aphi / V2-plunge diagnostics are recomputed from the interpolated tensor to match); facets outside the CSM convex hull fall back to nearest neighbour (zero on this fault — it is fully inside the CSM coverage).

```
--sample {nearest,linear}   (default nearest, reproducible/blocky;
                             linear = smooth orientation field)
```

Caveats. (1) Smoothing the magnitude scale is unaffected: Sv_eff is already depth-interpolated and R only sets the intermediate axis; `--sample` changes only the ORIENTATION continuity. (2) The mu cap is preserved under either sampling: for closure C1 the ratio k bounds mu <= (k-1)/(2 sqrt k) for ANY orientation, so interpolation only redistributes mu within [0, that bound]. (3) Linear interpolation smooths ACROSS the San Gorgonio style transition (transtension <-> transpression); it interpolates the tensor components (not the discrete Aphi regime), and the `csm_Aphi_cell` / `csm_eig_gap_cell` outputs flag where a smoothed facet straddles a style boundary.

---

6. Step 5 – Implementation and validation plan

Implementation (single file change, project_csm_stress_to_vtu.py)

1. Add magnitude model: `--magnitude-mode {constant, depth-R}`. `constant` keeps today's behavior (sigma1/2/3 flags) for reproducibility of the existing artifact.
2. `depth-R` mode flags, with rev-5 defaults shown: `--closure diff` (default; `ratio` for C1), `--diff-csv PATH` (default = the Luttrell file in `raw_data/luttrell_differential_stress/`; closure C2-s), `--diff-MPa 40` (scalar fallback when no csv), `--k-ratio 3.12` (only used by `--closure ratio`), `--pp-model hydrostatic` (default), `--density-source muscal` (default; `constant` + `--rho-const-kgm3` for the check) with `--material-nc PATH` (default `safs_seisol_v2_0_0_RSSRW/safs_material_cvm.nc`), `--axes csm` (default; `andersonian`

alt), `--sample nearest` (default; `linear` = interpolated/smooth orientation, see Section 5), optional `--taper-zseis-km` (off by default).

3. New per-cell outputs: sig1/sig2/sig3_eff_MPa_cell, csm_R_cell, csm_Aphi_cell, csm_V2_plunge_deg_cell, Sv_total/Sv_eff/P_p_MPa_cell, depth_m_cell, and (closure diff) D_diff_MPa_cell (the sampled differential, column 16 for C2-s), D_applied_MPa_cell (after the frictional cap), and a per-facet capped flag; record the capped-facet count in the summary.
4. Summary JSON: record every parameter + closure formulas used.

Validation checklist (all must pass before handing fields to easi)

- ☐ Shape recovery: eigenvalues of every built tensor reproduce the input R to ~1e-3 (closed-form, so failure = code bug).
- ☐ Ordering + positivity: sig1 >= sig2 >= sig3 > 0 on 100% of facets (count capped facets in C2 mode).
- ☐ SHmax of built tensor vs csv column 10 (median < 1 deg, as in the constant-magnitude run).
- ☐ Option A only: sigma_zz_eff vs Sv_eff(z) deviation stats; correlate with V2 plunge.
- ☐ SAF weak-fault sanity: distribution of resolved tau on the fault vs the ~20 MPa-class heat-flow bound (Lachenbruch & Sass 1980, 1992); mu_apparent median in the ~0.1–0.3 range implied by the high SHmax-to-fault angle (Hardebeck & Hauksson 2001; Townend & Zoback 2004). If violated, revisit k / D per Section 3.
- ☐ Luttrell self-consistency (closure C2-s, data now in hand): D_applied(x) == D_luttrell(x) on every un-capped facet; report the capped-facet count and the depths at which capping occurred (expected: shallow + San Gorgonio high-D facets). The Luttrell nn-sampling distance must stay within the grid spacing (~2 km), like the YHSM orientation sampler.
- ☐ Hypocenter facet report (same probe point as previous runs: 606971, 3707270, -4965.62).
- ☐ Diff against the constant-magnitude artifact (`csm_yhsm2013_stress_on_safs_mesh_*`): same orientation fields, magnitudes now depth-dependent.

---

7. References (exact citations; verification status as of 2026-06-12)

Group 1 — SAF system / southern California observations (these carry ALL parameter values). Verified online (publisher page) unless noted:

1. Yang, W. & Hauksson, E. (2013). The tectonic crustal stress field and style of faulting along the Pacific North America Plate boundary in Southern California. Geophysical Journal International 194(1), 100–117. doi:10.1093/gji/ggt113. [Source model YHSM-2013: orientations, phi/R/Aphi, SHmax.]
2. Zoback, M.D. & Healy, J.H. (1992). In situ stress measurements to 3.5 km depth in the Cajon Pass scientific research borehole: implications for the mechanics of crustal faulting. Journal of Geophysical Research 97(B4), 5039–5057. doi:10.1029/91JB02175. [Borehole 4.2 km from the southern SAF: mu = 0.6–1.0 frictional equilibrium with HYDROSTATIC pore pressure -> k and lambda values.]
3. Coyle, B.J. & Zoback, M.D. (1988). In situ permeability and fluid pressure measurements at ~2 km depth in the Cajon Pass research well. Geophysical Research Letters 15(9), 1029–1032. doi:10.1029/GL015i009p01029. [DIRECT fluid-pressure measurement near the southern SAF: ~5 % above hydrostatic — the key datum behind –pp-model hydrostatic.]

4. Fulton, P.M. & Saffer, D.M. (2009). Potential role of mantle-derived fluids in weakening the San Andreas Fault. Journal of Geophysical Research 114, B07408. doi:10.1029/2008JB006087. [SAF-specific pore-pressure DISTRIBUTION model: overpressure localized in the fault zone (~1 MPa @ 2.7 km, ~17 MPa @ 6 km excess), country rock near-hydrostatic; supports hydrostatic regional lambda + optional fault-local correction.]
5. Townend, J. & Zoback, M.D. (2004). Regional tectonic stress near the San Andreas fault in central and southern California. Geophysical Research Letters 31, L15S11. doi:10.1029/2003GL018918. [Strong crust / weak SAF; SHmax at high angle to the fault.]
6. Hardebeck, J.L. & Hauksson, E. (2001). Crustal stress field in southern California and its implications for fault mechanics. Journal of Geophysical Research 106(B10), 21859–21882. doi:10.1029/2001JB000292. [Stress rotations near the SAF; low apparent fault friction; mu_apparent validation target.]
7. Luttrell, K. & Smith-Konter, B. (2017). Limits on crustal differential stress in southern California from topography and earthquake focal mechanisms. Geophysical Journal International 211(1), 472–482. doi:10.1093/gji/ggx301. (SCEC contribution #6064.) [Closure C2/C2-s, the rev-5 DEFAULT: D >= ~20 MPa regionally, ~62 MPa near rugged SAF topography. The gridded model has been OBTAINED from the SCEC CSM portal as `CSM_data_1781537253009.csv` (CSM model "Luttrell-2017", placed in `raw_data/luttrell_differential_str` 35-column tension-positive export, differential stress in column 16, single depth 5 km, depth-invariant, ~0.02 deg grid covering the whole SAFS fault footprint. Used as the spatial D(x,y) field.]
8. Lachenbruch, A.H. & Sass, J.H. (1980). Heat flow and energetics of the San Andreas fault zone. Journal of Geophysical Research 85(B11), 6185–6222. doi:10.1029/JB085iB11p06185. [Heat-flow bound: SAF-resolved shear <= ~20 MPa; validation.]
9. Lachenbruch, A.H. & Sass, J.H. (1992). Heat flow from Cajon Pass, fault strength, and tectonic implications. Journal of Geophysical Research 97(B4), 4995–5030. doi:10.1029/91JB01506. [Same bound specifically at Cajon Pass, i.e. the southern SAF; pages re-verify before manuscript use.]
10. Zoback, M.D. et al. (1987). New evidence on the state of stress of the San Andreas fault system. Science 238, 1105–1111. [Weak-SAF / strong-crust synthesis.]
11. Scholz, C.H. (2000). Evidence for a strong San Andreas fault. Geology 28(2), 163–166. doi:10.1130/0091-7613(2000)28<163:EFASSA>2.0.CO;2. [Strong-fault counterpoint; bounds the debate the validation checklist navigates.]
12. Nazareth, J.J. & Hauksson, E. (2004). The seismogenic thickness of the southern California crust. Bulletin of the Seismological Society of America 94(3), 940–960. [z_seis = 15.0 +1.2/-1.1 km average, < 10 km Salton Trough; deep taper depth scale.]
13. Kennedy, B.M. et al. (1997). Mantle fluids in the San Andreas fault system, California. Science 278, 1278–1281. [Mantle-helium evidence feeding the Fulton & Saffer model; standard citation, re-verify pages before manuscript use.]

Group 2 — setting-independent mechanics and data relations (no event-specific values):

14. Hubbert, M.K. & Rubey, W.W. (1959). Role of fluid pressure in mechanics of overthrust faulting, I. GSA Bulletin 70, 115–166. [Effective-stress principle + lambda notation only; verified.]
15. Anderson, E.M. (1951). The Dynamics of Faulting and Dyke Formation with Applications to Britain, 2nd ed., Oliver & Boyd. [Andersonian regimes: one principal axis vertical; classic

monograph.]
16. Byerlee, J. (1978). Friction of rocks. Pure and Applied Geophysics 116, 615–626. [Laboratory mu = 0.6–1.0 range quoted by the borehole studies; standard citation, re-verify pages before manuscript use.]
17. Brocher, T.M. (2005). Empirical relations between elastic wavespeeds and density in the Earth's crust. Bulletin of the Seismological Society of America 95(6), 2081–2092. doi:10.1785/0120050077. [The conventional empirical Vp->rho relation for crustal models, calibrated on saturated in-situ rocks (-> bulk density). Whether MUSCAL used exactly this relation is TO CONFIRM, Section 2.1; citation itself verified.]

Group 3 — definitions used by the csv columns (cited by the csv header itself; standard citations, re-verify volume/pages before manuscript use):

18. Gephart, J.W. & Forsyth, D.W. (1984). An improved method for determining the regional stress tensor using earthquake focal mechanism data… Journal of Geophysical Research 89(B11), 9305–9320. [csv R, column 13.]
19. Angelier, J. (1979). Determination of the mean principal directions of stresses for a given fault population. Tectonophysics 56, T17–T26. [csv phi, column 12.]
20. Simpson, R.W. (1997). Quantifying Anderson's fault types. Journal of Geophysical Research 102(B8), 17909–17919. [csv Aphi, col 14.]

Group 4 — material-model provenance (the density column of Section 2.1):

21. Yeh, T.-Y., Ben-Zion, Y. & Olsen, K.B. (2026). Multi-Scale P and S Seismic Velocity Models of California and Western Nevada. ESSOAr preprint, doi:10.22541/essoar.15001929/v1. [PREPRINT, not yet peer-reviewed — the MUSCAL model behind `safs_material_cvm.nc` (CVM Explorer abbr "muscal", UCVM component `SCECcode/muscal`, authors Yeh & Ben-Zion). Confirm the density-assignment recipe here; replace with the peer-reviewed citation when it appears.]
22. Doody, C., Rodgers, A., Afanasiev, M., Boehm, C., Krischer, L., Chiang, A. & Simmons, N. (2023). CANVAS: an adjoint waveform tomography model of California and Nevada. Journal of Geophysical Research: Solid Earth 128(12), e2023JB027583. doi:10.1029/2023JB027583. [Base model MUSCAL builds on; verified.]

Group 5 — frictional-strength cap on the differential stress (the near-surface cap of closure C2, rev 6; Byerlee 1978 = ref 16 and Zoback & Healy 1992 = ref 2 above also carry this claim). Verified online (publisher page) 2026-06-15:

23. Sibson, R.H. (1974). Frictional constraints on thrust, wrench and normal faults. Nature 249(5457), 542–544. doi:10.1038/249542a0. [Setting-independent mechanics: the frictional differential-stress limit per Andersonian regime increases ~linearly with depth from ~0 at the free surface — supports D_max proportional to Sv_eff.]
24. Brace, W.F. & Kohlstedt, D.L. (1980). Limits on lithospheric stress imposed by laboratory experiments. Journal of Geophysical Research 85(B11), 6248–6252. doi:10.1029/JB085iB11p06248. [Setting- independent: the brittle crust's differential stress is bounded by Byerlee frictional strength; "a good upper or lower bound to observed in-situ stresses … for pore pressure hydrostatic" — the direct justification for capping D at the frictional ceiling.]
25. Townend, J. & Zoback, M.D. (2000). How faulting keeps the crust strong. Geology 28(5), 399–

402. doi:10.1130/0091-7613(2000)028<0399:HFKTCS>2.3.CO;2. [SAF / central-southern California: the crust is maintained in frictional- failure equilibrium (Byerlee-bounded) — the cap ceiling is where the crust actually sits, not merely an upper bound. Companion to Townend & Zoback 2004 = ref 5.]

EXCLUDED (Parkfield / creeping central SAF — user directive 2026-06-12; listed so they are not re-introduced by accident): Hickman & Zoback (2004) GRL 31 L15S12 (SAFOD pilot-hole stress magnitudes); Lockner et al. (2011) Nature 472 82–85 (SAFOD core gouge friction).

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8. Decisions

RESOLVED by user directive 2026-06-15 (rev 5)

1. Closure default: **C2-s** — the spatial Luttrell differential-stress field D(x,y) (Section 3 C2-s, Section 4). Flips the rev-3 recommendation; C1 ratio k = 3.12 retained as `--closure ratio`.
2. Pore pressure: **hydrostatic** regional default (Coyle & Zoback 1988; Zoback & Healy 1992; Fulton & Saffer 2009). Fault-local excess-P_p (Fulton & Saffer profile) deferred to the friction setup.
3. Density: **MUSCAL** lateral-mean rho(z) profile (Section 2.1); `constant` kept only for the magnitude check.
4. C2-s data acquisition: **DONE** — the Luttrell-2017 CSM-portal export is in `raw_data/luttrell_differential_` (same 35-column format as the YHSM csv; D = column 16).
5. MUSCAL density recipe: **downgraded to a non-blocking manuscript footnote** — use the shipped rho as-is; confirm the Vp->rho relation from the Yeh, Ben-Zion & Olsen preprint before publication, not before implementation.

STILL OPEN

3. Axes: Option A (full CSM eigenvectors, keeps San Gorgonio transpression) as default vs Option B (strict Andersonian). Plan recommends A.
4. Deep taper: off by default; if enabled, z_seis = 15 km (Nazareth & Hauksson 2004) or a locally varying map later?